LEHD Snapshot Documentation, Release S2021_R2022Q4

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Abstract

The Longitudinal Employer-Household Dynamics (LEHD) data at the U.S. Census Bureau is a quarterly database of linked employer-employee data covering over 95% of employment in the United States. These data are used to produce a number of public-use tabulations and tools, including the Quarterly Workforce Indicators (QWI), LEHD Origin-Destination Employment Statistics (LODES), Job-to-Job Flows (J2J), and Post-Secondary Employment Outcomes (PSEO) data products. Researchers on approved projects may also access the underlying LEHD microdata directly, in the form of the LEHD Snapshot restricted-use data product. This document provides a detailed overview of the LEHD Snapshot as of release S2021_R2022Q4, including user guidance, variable codebooks, and an overview of the approvals needed to obtain access. Updates to the documentation for this and future snapshot releases will be made available in HTML format on the LEHD website.

* This research describes data from the Census Bureau's Longitudinal Employer Household Dynamics Program, the original creation of which was partially supported by the following National Science Foundation (NSF) Grants SES-9978093, SES-0339191 and ITR0427889; National Institute on Aging Grant AG018854; and grants from the Alfred P. Sloan Foundation. Additionally, the current authors acknowledge the extensive contribution over the years by many, many individuals to the cumulative knowledge reflected in this document, too many to adequately enumerate here. This document has been reviewed to ensure that no confidential information is disclosed.

The latest version of this documentation is published in HTML format, and is publicly available at https://lehd.ces.census.gov/data/lehd-snapshot-doc/latest/

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CHAPTER

ONE

INTRODUCTION

The Longitudinal Employer-Household Dynamics (*LEHD*) data at the U.S. Census Bureau is a quarterly database of linked employer-employee data covering over 95% of employment in the United States.[1] The *LEHD* data are generated by merging previously collected survey and administrative data on jobs, businesses, and workers. By integrating administrative data with existing census and surveys, a national longitudinal jobs database for the U.S. is generated at very low cost and with no additional respondent burden. The LEHD microdata is available to approved researchers in the Federal Statistical Research Data Center (FSRDC) network. LEHD data is also available to the public in tabular form in the Census Bureau's Quarterly Workforce Indicators (*QWI*), LEHD Origin-Destination Employment Statistics (*LODES*), Job-to-Job Flows (*J2J*), and the Post-Secondary Employment Outcomes (*PSEO*) data.

In order to facilitate researcher use of the microdata, LEHD infrastructure files are periodically collected into a standardized form known as the LEHD Snapshot, which is then released into the FSRDC network. The purpose of this documentation and codebook is to provide a limited overview of the components of the LEHD Snapshot files, as well as detailed codebooks for researcher use. For an overview of the LEHD infrastructure, see Abowd *et al.* [2009]. This documentation is not intended to provide details of specific LEHD data products, or of the disclosure limitation measures used to produce those products. For more information on specific LEHD data products and their disclosure limitation techniques, see Abowd *et al.* [2006], Haney *et al.* [2017], Machanavajjhala *et al.* [2008] .

The LEHD Snapshot consists almost exclusively of restricted-use administrative microdata. Therefore, as with all projects using restricted data, any projects using the LEHD Snapshot must undergo standard review and approval processes. Researchers are required to demonstrate that they cannot conduct their research using public-use data sources, and they must identify the ways in which their project will benefit both the objectives of the U.S. Census Bureau and the broader research community. Before requesting access to the LEHD Snapshot, researchers are encouraged to examine the public data offerings of the LEHD program at https://lehd.ces.census.gov/data/, as well as the broader public offerings at https://data.census.gov/. For more details on the FSRDC program and the requirements for project approval, see the FSRDC about page as well as the *approval guide below*.

This documentation is current as of LEHD Snapshot release S2021_R2022Q4, which contains available data through 2021. Features may change in subsequent snapshot releases. The latest version of this documentation is published in HTML format, and is publicly available at https://lehd.ces.census.gov/data/lehd-snapshot-doc/latest/. For details on how to cite the data and this documentation, *see below*. Documentation on the 2004, 2008, and 2014 snapshot releases is also available in working paper format [McKinney and Vilhuber, 2011, McKinney and Vilhuber, 2011, Vilhuber, 2018].

[1] This research describes data from the Census Bureau's Longitudinal Employer Household Dynamics Program, the original creation of which was partially supported by the following National Science Foundation (NSF) Grants SES-9978093, SES-0339191 and ITR0427889; National Institute on Aging Grant AG018854; and grants from the Alfred P. Sloan Foundation. Additionally, the current authors acknowledge the extensive contribution over the years by many, many individuals to the cumulative knowledge reflected in this document, too many to adequately enumerate here. This document has been reviewed to ensure that no confidential information is disclosed.

1.1 Earnings Coverage

Covered private-industry employment in the LEHD data includes most corporate officials, all executives, all supervisory personnel, all professionals, all clerical workers, many farmworkers, all wage earners, all pieceworkers, and all part-time workers. Workers on paid sick leave, paid holiday, paid vacation, and the like are also covered. Workers on the payroll of more than one firm during the period are counted by each employer that is subject to *UI*, as long as those workers satisfy the preceding definition of employment. Workers have UI wages filed in every quarter they are covered, even though their wages may not be subject to UI tax in the latter months of the year.

Notable exclusions from UI coverage among private sector employers are independent contractors, the unincorporated self-employed, railroad workers covered by the railroad unemployment insurance system, some family employees of family-owned businesses, certain farm workers, students working for universities under certain cooperative programs, salespersons primarily paid on commission, and workers of some non-profits. States have some leeway in designating coverage, for a complete list, see the coverage section of the most recent Comparison of State UI Laws.

Covered employment in the LEHD data includes most employees of state and local governments with the exception of elected officials, members of a legislative body or members of the judiciary, and some emergency employees.

Federal government workers are not covered by state UI and do not appear in UI wage record data.

1.2 LEHD Confidential Microdata: Job, Employer, and Person-Level Files

The LEHD Snapshot is organized into sets of tables providing job-level, employer-level, and person-level data. The scope of each table is either national (all states) or a single state. National tables will have a *US* in the table name, while state tables will have the two-character state postal code. For purposes of this documentation, the postal code will be replaced with *ZZ*. Tables that are national in scope will not have the complete panel of states in all quarters because of limited availability of state job data both at the beginning and end of the time series. For more details, see *Availability of Data*.

1.2.1 Job-Level Files

Job-level data provide earnings for jobs, which is a link between a worker and a firm. The employment coverage in the LEHD job-level microdata files described below is UI-covered employment only (see *Earnings Coverage* for details on covered employment and wages). Researchers interested in federal jobs should request the (*OPM*) microdata separately as part of their research request.

Table 1: Job-Level Files in the LEHD Snapshot

Table	Description	Scope	Key
Earnings Availability (EHF_ALL_AVAILABILITY)	Indicates data availability ranges for all states.	National	State
Earnings Indicators (EHF_US_INDICATORS)	Indicates whether or not a PIK had any earnings reported in any state in the LEHD system.	National	PIK YEAR
Employment History File (EHF_ZZ)	Quarterly earnings for each job in a state as reported by the employer to the state's UI system.	State	PIK SEIN SEINUNIT YEAR
Job History File (JHF_ZZ)	Wide version of earnings history that contains imputed establishment (SEINUNITs) for multi-unit firms, and a job-level ID useful for tracking job spells across employer identifier changes.	State	PIK SEIN SPELL_U2W

1.2.2 Employer-Level Files

Employer-level data contain characteristics of firms, as well as summary measures calculated from job data.

Table 2: Employer-Level Files in the LEHD Snapshot

Table	Description	Scope	Key
Employer Characteristics File, Firm (ECF_ZZ_SEIN)	Sourced from the SEINUNIT file, this file rolls up employer characteristics to the SEIN level.	State	SEIN YEAR QUARTER
Employer Characteristics File, Establishment (ECF_ZZ_SEINUNIT)	An establishment level file with industry, size, and location.	State	SEIN YEAR QUARTER SEINUNIT
Employer Characteristics File, National Firm (ECF_ZZ_SEIN_T26)	Contains the national firm age and size data for employers in the ECF. This file requires IRS approval.	State	SEIN YEAR QUARTER
Quarterly Workforce Indicators (QWI_ZZ_SEINUNIT)	The microdata version of the public use QWI data, containing establishment-level hires, separations, net job growth, and average earnings.	State	SEIN YEAR QUARTER SEINUNIT
Successor-Predecessor File (SPF_ZZ)	Flows of separations and hires between employers (SEINs) used to identify successor-predecessor relationships.	State	SEIN SEIN_SUCC QTIME

1.2.3 Person-Level Files

Person-level tables provide demographic characteristics of individuals and observed residence geography.

Table 3: Person-Level Files in the LEHD Snapshot

Table	Description	Scope	Key
Individual Characteristics File (ICF_US)	Demographics, place of birth, and education of workers.	National	PIK
Date of Birth/Sex/Place of Birth Implicates (ICF_US_IMPLICATES_AGE_SEX_POB)	Multiply imputed date of birth, sex, and place of birth variables to complete missing information.	National	PIK
Education Implicates (ICF_US_IMPLICATES_EDUCATION)	Multiply imputed education variables to complete missing information.	National	PIK
Race/Ethnicity Implicates (ICF_US_IMPLICATES_RACE_ETHNICITY)	Multiply imputed race and ethnicity variables to complete missing information.	National	PIK
Residence Geography, 1999-2011 (ICF_US_RESIDENCE_CPR)	Residence geography prior to 2012 for individuals found in the wage data.	National	PIK ADDRESS_YEAR
Residence Geography, 2012 forward (ICF_US_RESIDENCE_RCF)	Residence geography from 2012 forward for individuals found in the wage data.	National	PIK ADDRESS_YEAR

1.3 Linking the job, employer, and person data

- To link worker demographics to the jobs data, link the EHF or JHF to the ICF US via the PIK.
- To attach the characteristics of the employer (SEIN) to the job, link the EHF or JHF to the ECF SEIN using the SEIN.
- To attach the characteristics of the establishment to the job, use the first implicate of the worker-to-establishment imputation (SEINUNIT1) on the JHF to link to the ECF SEINUNIT table.
  - All implicates (SEINUNIT1-SEINUNIT10) can be used to produce standard errors that account for imputation variability [McKinney *et al.*, 2021].
  - Because of longitudinal restrictions on establishment imputation, jobs may be active at establishments in quarters that the establishment does not appear on the ECF. See *Adding Establishment Characteristics* for more information.

1.4 Approvals Needed for FSRDC Research

In addition to Census data, the administrative records that combine to form the LEHD infrastructure derive from several different federal and state agencies. Approval of a proposed FSRDC project by other agencies may be required, as follows:

- Requests for state-level tables will follow the procedure specified by the Memorandum of Understanding (MOU) with that state. Most commonly, the state agency is given the opportunity to approve proposed projects.
- Person-level demographic characteristics (ICF) require the approval of the Social Security Administration (SSA).

- Employer-level Title 26 (T26) data requires the additional approval of the Internal Revenue Service (IRS). This includes the Federal EIN, as well as age and size of the national firm.

The codebook for each table will indicate which approvals are required.

The job-level tables largely consist of state-provided data and so go through the LEHD state data owner approval process. The indicator and availability files do not require state approval. There is no federal tax information or Social Security administrative data in any of the job-level tables, so no additional IRS or SSA approvals are required.

1.5 Snapshot Packages in the FSRDC

LEHD Snapshot tables are delivered in packages according to the level of data in the tables. Tables that require additional approvals are provided in separate packages.

Table 4: Snapshot Packages in the FSRDC

Package	Tables
Jobs Data	EHF_ZZ JHF_ZZ EHF_US_INDICATORS EHF_ALL_AVAILABILITY
Employer Data (non-T26)	ECF_ZZ_SEIN ECF_ZZ_SEINUNIT SPF_ZZ QWI_ZZ_SEINUNIT
Employer Data (T26)	ECF_ZZ_SEIN_T26
Person Demographics (Workers Only)	ICF_US ICF_US_IMPLICATES_AGE_SEX_POB ICF_US_IMPLICATES_RACE_ETHNICITY ICF_US_IMPLICATES_EDUCATION
Person Residence (Workers Only)	ICF_US_RESIDENCE_CPR ICF_US_RESIDENCE_RCF

1.6 Quarter Indexing - *qtime*

On several LEHD tables, the year and quarter is replaced by or supplemented with an index for ease of counting quarters in longitudinal references. This is initialized with 1985Q1 being set to 1. The following SAS formulae can be used to switch back and forth:

- `qtime = (year-1985) * 4 + quarter`
- `year = int((qtime-1)/4) + 1985`
- `quarter = mod(qtime-1, 4) + 1`

1.7 Availability of Data

The following table indicates the availability of LEHD data by state for the EHF, ECF, and QWI tables. This includes all states that are or have been members of the LED partnership. In general, LEHD snapshot release S2021_R2022Q4 contains all available data through 2021.

- The EHF range represents all of the quarters of UI wage data that have been received.
- The QWI range may begin later than the EHF range if certain data quarters do not meet quality standards for publication in LED data products.
  - Earnings quarters on the JHF are restricted to the QWI range.
- The ECF range may start before the EHF range if employer-level data submissions (QCEW) are available prior to the beginning of the UI wage data series.
- End dates will usually be the same for all tables within a state. States that are not active partners may not be available for the latest quarters.

Table 5: Available Quarters of LEHD Data

State	ECF Start	EHF Start	QWI/JHF Start
Alabama	2001Q1	2001Q1	2001Q1
Alaska	1990Q1	1990Q1	2000Q1
Arizona	1992Q1	1992Q1	2004Q1
Arkansas	2002Q3	2002Q3	2002Q3
California	1991Q1	1991Q3	1991Q3
Colorado	1990Q1	1990Q1	1993Q2
Connecticut	1996Q1	1996Q1	1996Q1
Delaware	1997Q1	1998Q3	1998Q3
District of Columbia	2000Q4	2002Q2	2005Q2
Florida	1989Q1	1992Q4	1997Q4
Georgia	1994Q1	1994Q1	1998Q1
Hawaii	1995Q4	1995Q4	1995Q4
Idaho	1990Q1	1990Q1	1991Q1
Illinois	1990Q1	1990Q1	1993Q2
Indiana	1990Q1	1990Q1	1998Q1
Iowa	1990Q1	1998Q4	1998Q4
Kansas	1990Q1	1990Q1	1993Q1
Kentucky	1996Q4	1996Q4	2001Q1
Louisiana	1990Q1	1990Q1	1995Q1
Maine	1996Q1	1996Q1	1996Q2
Maryland	1985Q4	1985Q4	1990Q1
Massachusetts	2002Q1	2002Q1	2010Q1
Michigan	1998Q1	1998Q1	2000Q3
Minnesota	1994Q3	1994Q3	1994Q3
Mississippi	2003Q3	2003Q3	2003Q3
Missouri	1990Q1	1990Q1	1995Q1
Montana	1993Q1	1993Q1	1993Q1
Nebraska	1999Q1	1999Q1	1999Q1
Nevada	1998Q1	1998Q1	1998Q1
New Hampshire	2003Q1	2003Q1	2003Q1
New Jersey	1995Q1	1996Q1	1996Q1
New Mexico	1990Q1	1995Q3	1995Q3

continues on next page

Table 5 – continued from previous page

State	ECF Start	EHF Start	QWI/JHF Start
New York	1990Q1	1995Q1	2000Q1
North Carolina	1990Q1	1991Q1	1992Q4
North Dakota	1998Q1	1998Q1	1998Q1
Ohio	1994Q1	1994Q1	2000Q1
Oklahoma	1999Q1	2000Q1	2000Q1
Oregon	1990Q1	1991Q1	1991Q1
Pennsylvania	1991Q1	1991Q1	1997Q1
Rhode Island	1990Q1	1995Q1	1995Q1
South Carolina	1998Q1	1998Q1	1998Q1
South Dakota	1994Q1	1994Q1	1998Q1
Tennessee	1998Q1	1998Q1	1998Q1
Texas	1990Q1	1995Q1	1995Q1
Utah	1990Q1	1999Q1	1999Q3
Vermont	2000Q1	2000Q1	2000Q1
Virginia	1995Q3	1998Q1	1998Q3
Washington	1990Q1	1990Q1	1990Q1
West Virginia	1990Q1	1997Q1	1997Q1
Wisconsin	1990Q1	1990Q1	1990Q1
Wyoming	1992Q1	1992Q1	2001Q1

1.8 Citing the Data, Sponsors, and Documentation

1.8.1 Acknowledging the Sponsors

The LEHD Snapshot draws on a data infrastructure that received substantial funding from a number of funding agencies and foundations. We strongly encourage researchers to acknowledge that funding in their paper's "Acknowledgements" or data appendix. The following statement may be used:

> This research uses data from the Census Bureau's Longitudinal Employer-Household Dynamics program, which was partially supported by the following National Science Foundation grants SAS-9978093, SES-0339191, and ITR-0427889; National Institute on Aging grant AG018854; and grants from the Alfred P. Sloan Foundation.

1.8.2 Citing the Data

The data may be cited as follows:

Suggested citation

- U.S. Census Bureau. LEHD Snapshot Release S2021. [Computer file], U.S. Census Bureau, Center for Economic Studies, Research Data Centers [distributor], Washington, DC, 2022.

1.8.3 Data Access Statement

Many journals have adopted stringent data availability requirements. Researchers will need to work with the Census Bureau in order to ensure availability of their programs and research extracts. A statement similar to the following has been successfully used for accepted papers:

> The data used for this paper were prepared in the U.S. Census Bureau's secure computing facilities under an authorized project using the Federal Statistical Research Data Center network. The exact analysis files have been fully archived so that the programming sequence submitted in compliance with the [JOURNAL]'s editorial policy can be run in its entirety, except for the component that extracts the analysis sample from the underlying confidential databases. I grant any researchers with appropriate Census-approved project permission to use my exact research files provided that those files were among the ones that they requested when the approval was obtained (a Census Bureau requirement). In compliance with the [JOURNAL]'s editorial policy, I am submitting the list of those files, and the last known location of the archive on the Census Bureau's RDC network as of [date]. I authorize the editorial staff of the [JOURNAL] to release this list and my statement of cooperation to any researcher who requests it, as well as to the U.S. Census Bureau or any agency cooperating with the Census Bureau in supervising research that uses the restricted-access data that I have used.

CHAPTER

TWO

JOBS-LEVEL FILES

2.1 Employment History Files

2.1.1 Overview

The Employment History Files are a set of job-level tables in the LEHD data. The source of the jobs data is earnings records collected by state unemployment insurance programs, called *UI* wage record data. These consist of quarterly reports of earnings for all UI-covered employees, provided by employers to states and shared with Census via a cooperative partnership. More information on UI-covered employment and wages is available in previous sections of this document.

The *EHF* file family consists of four tables:

- **Employment History File (EHF_ZZ)**

  This table contains annual records with quarterly earnings for each covered job in a state. This is a simple file that is relatively easy to use for research analysis.

  - Scope: State
  - Key: PIK SEIN SEINUNIT YEAR

- **Job History File (JHF_ZZ)**

  This is a wide file containing the earnings arrayed in quarterly variables for each job. In addition to the earnings available in the EHF, the JHF contains imputed establishments for multi-unit employers and successor-predecessor information. The earnings arrays in the JHF are also useful for calculating worker flows. Job spells may be recorded on multiple records in this table as a result of identifier changes or imputation.

  - Scope: State
  - Key: PIK SEIN SPELL_U2W

- **Earnings Indicators (EHF_US_INDICATORS)**

  This file contains flags indicating whether or not a PIK had earnings reported in any state in the LEHD system. This is useful for identifying bias in research samples using a subset of state UI data.

  - Scope: National
  - Key: PIK YEAR

- **Earnings Availability (EHF_US_AVAILABILITY)**

  This file provides quarter ranges for which state data is available in the EHF and JHF tables. This is useful for identifying bias in research samples.

  - Scope: National

  - Key: STATE

Researchers are encouraged to request all four of the job-level files described for analysis. The most useful file for most researchers is the JHF. However, the EHF can be useful for analyses where establishment-level characteristics or the longitudinal nature of the job is less important than person-level earnings. The last indicators and availabilitiy tables are useful for situations where sample bias or left-censoring of job histories might be an issue.

2.1.2 User Guidance

Identifiers: Employers vs. Firms vs. Establishments

UI wage records are reported at the state tax identifier level (SEIN) for all states - except for Minnesota where they are reported at the establishment level (SEINUNIT). We typically refer to SEINs as *employers* in this document, but users should be aware they are fundamentally a tax entity and distinct from an establishment or a firm. While identifiers that correspond to the definition of a firm as used by the Longitudinal Business Database (LBD) are available in the *ECF T26 file*, job-level and employer-level files are generally linked at either the employer or establishment level.

For further details on the definition and usage of each identifier, see the *ECF documentation*.

For further details on our establishment imputation methodology, see sections *below*.

Counting Jobs and Constructing Longitudinal Job Histories on the JHF

On the JHF, job spells are arrayed wide by quarter using *qtime* indexing. There can be one or many records per PIK-SEIN, depending on firm structure and information availability.

- *Single establishment employer (SEIN)*
  - There is one record per PIK-SEIN for these jobs. The establishment will be 00000.
- *Multi establishment employer (SEIN) with impute*
  - The PIK-SEIN record will have ten imputes of the establishment attached to it.
  - In cases where a worker leaves an employer for more than a year or the establishment structure changes significantly (births/deaths), the JHF record may be broken into multiple records by quarter ranges.
  - The establishment impute is performed independently for each record on the JHF.
- *Multi establishment employer (SEIN) with direct reporting (MN only)*
  - There will be one record for each PIK-SEIN-SEINUNIT when there is regular reporting of the establishment on the UI record.
  - Each PIK-SEIN-SEINUNIT record is considered a separate longitudinal job for purposes of QWI calculation.
  - In cases of irregular establishment reporting patterns on the wage records, SEINs can have jobs with directly reported establishments in some quarters and imputes in other quarters, or imputes in all quarters.

The SPELL_U2W variable is created as a simple counter within each PIK-SEIN to form a key that can uniquely identify each record on the JHF. Hence, the key on the table is PIK-SEIN-SPELL_U2W. However, a single JHF record may not be a complete longitudinal history of a job spell. In order to properly link the JHF records to construct the full job spell, the longitudinal job identifier (FID) is added to the JHF. This variable is created as follows:

- A sequential counter is created for all JHF records for a single PIK.
- All jobs in a set that is considered part of the same longitudinal history are assigned the lowest value of the counter within the set.

- The FID is created by turning the counter into a character field prepending with the state numeric FIPS code to make it nationally unique within a PIK.

The PIK-FID should be considered the unit of analysis when constructing the longitudinal history of the job spell. The job can be considered active in any quarter that the PIK-FID has positive earnings on any JHF record. Note that earnings can be reported on multiple JHF records in the same quarter within a PIK-FID group, and these should be summed. For QWI calculation purposes, counts are weighted by the share of total earnings for the PIK-FID, and divided by 10 if multiple imputation is used. This method ensures that each active PIK-FID will be counted as one job in QWI measures.

Accounting for Successor-Predecessor Events

A *job* in the EHF is a PIK-SEIN pair with positive earnings in a quarter. However, a worker's *job spell* can be broken up over multiple SEINs (i.e., be multiple rows in the JHF) if the employer SEIN changes during the spell. Employer tax identifiers change quite frequently in the LEHD data, usually due to firm restructuring. There are also limited instances where virtually all SEINs change within a state in a single quarter. If using the job-level data to identify job starts, separations, or job tenure, successor-predecessor (S-P) information to link jobs must be used to avoid bias arising from firm S-P events. S-P information must also be used to avoid double counting of jobs when earnings are provided by multiple related SEINs in the same quarter.

The SPF reports all significant flows of workers between SEINs. A subset of these flows have been identified as representing S-P events according to business rules described in *Identifying Transitions in UI Data*. Workers that traverse between two SEINs engaged in a S-P relationship should be considered as having a continuous job history across SEINs. In addition, if the worker received earnings from both SEINs in the same quarter, a common feature of an S-P event, the earnings should be combined, not counted as two separate jobs.

In the JHF, records in S-P relationships are linked by assigning the same FID, as described above. The mechanics of linking records follows:

- All separations and accessions are identified from all records for a PIK within the JHF.
  - A separation is a quarter of positive earnings followed by a quarter of no earnings, and an accession is the reverse.
  - Separations and accessions can occur in the beginning, middle, or end of the quarter range on the JHF record.
- A separation is linked with every accession in the same or following quarter to find candidates for linkage.
- If a S-P relationship identified on the SPF coincides with a link found for a PIK, the JHF records are joined via the FID.
  - Because significant flows have been observed between two SEINs in quarters proximate to the officially recognized transition quarter, links at the PIK level will be made if they are found within several quarters before or after the transition quarter.
  - Some links are provided via internal clerical tables, particularly to handle large-scale SEIN changes. These may not be included in the SPF, but will be incorporated into the FID.
- The linkage process works iteratively to link all related JHF records to the same FID.

Using the Worker-to-Establishment Imputes

The Census Bureau calculates establishment-level workforce statistics from the LEHD data by imputing establishments for jobs at multi-unit SEINs using multiple imputation. The 10 implicates from that imputation are available on the JHF for researchers. What follows is some general guidance on using the imputed values:

- First establish if you need the imputed establishment for your analysis. Many researchers use employment-weighted SEIN-level employer characteristics from the ECF. In many cases this might be a preferred approach if you are not using all 10 implicates.
- While we recommend using all 10 implicates, we are aware some researchers choose to use only one. If you choose to use only one of the imputes, we recommend you use the first implicate.

Worker-to-Establishment Methodology (U2W)

The full description of the worker-to-establishment imputation is available in Abowd *et al.* [2009], but we describe it briefly here. The model assigns workers to establishments using the relative size of the establishments and the distance between the worker's residence and the establishment. In other words, the model favors imputing workers to the closest establishment, but must also allocate more workers to larger establishments. Imputed values are drawn 10 times to better capture the uncertainty in the imputation. Model parameters for the imputation are determined using the Minnesota data, which does have SEINUNIT-level jobs data.

To give a very simple example, suppose worker Z works at a SEIN with 3 establishments, A, B and C. If the model determines there is a 60 percent likelihood Z works at A and 20 percent chance Z works at B or C, the 10 implicates will be: A, A, A, A, A, A, B, B, C, C. Multiple imputation gives the user more information about the uncertainty of the imputation than a single impute.

Adding Establishment Characteristics

Employer characteristics to attach to jobs can be retrieved from one of the following sources:

- **ECF_SEIN** table - Employment-weighted modal characteristics from establishment data
- **ECF_SEIN_T26** table - Firm level characteristics using Title 26 data (age/size)
- **ECF_SEINUNIT** table - Establishment level characteristics
- **QWI_SEINUNIT** table - Establishment level characteristics, with some longitudinal editing after linking with jobs data

Establishment-level characteristics include ownership, industry, and geography. Ownership will almost always be the same for all establishments in an SEIN, and cases where that is not true may suggest a data irregularity. There is significant establishment heterogeneity in substate georaphy, and less but still significant heterogeneity in industry. While SEIN-level characteristics may provide a first-order approximation, there will likely be mass points and redistributions in counts relative to establishment-level statistics.

One complication of using establishment characteristics, however, is that the establishment will not always be present in the ECF SEINUNIT or QWI SEINUNIT tables in all quarters that a job to which it has been imputed is active on the JHF. This is usually a function of the operation of the worker-to-establishment methodology. While establishment candidates are intended to be available for the whole of a worker's job spell at an SEIN, there are allowances for exceptions to this rule. There are several possible fallback strategies to handle these cases:

1. Use the SEIN-level modal characteristics. This is typically the most straightforward approach, but it can result in mass points and sudden longitudinal changes to the assignment.

2. Perform a longitudinal edit to get characteristics from the establishment in the nearest quarter it exists. This will maintain longitudinal consistency in assigned characteristics, but it is possible the establishment did not exist and using that geography (or industry) could be incorrect.
3. Draw an establishment from the set of establishments that do exist within the SEIN in the time period. This approach would come closest to replicating the distribution of the characteristics of the firm in the quarters of interest, though there may be minimal information on the job level to condition the impute. It is also likely the most complicated to implement.

While no particular approach is necessarily preferred, it is recommended that researchers applying establishment-level characteristics be aware of this phenomenon and apply a consistent approach to handling it.

Imputed Wage Records

About 1.5 percent of jobs are missing in the UI wage record data, jobs not reported by the employer or reported after the records are sent to Census Bureau. In these cases, wage records are imputed. A simple explanation of this imputation follows:

- A first stage identifies likely cases of nonreporting by identifying employers where the number of wage records reported in a quarter is less than the minimum expected given employment in the QCEW.
- In the second stage workers with jobs at the firm but no earnings in the underreported quarter form the candidate pool for imputation. Workers are imputed to active status in that quarter conditioning on the worker's empoloyment in the previous and next quarter using a posterior built from historical employment patterns. Generally speaking, workers with active jobs in the previous and following quarter have the highest probability of being imputed to active and receiving imputed earnings. Workers who are never observed working at the firm will never be imputed as working at the firm.

The Census Bureau currently fills only one-quarter reporting gaps, which is the most common type of reporting issue observed in the data. Employers with persistent reporting issues are a more difficult case for imputation.

Selecting a Random Subsample of Persons

The *RANDOM_PIK_GROUP* variable can be used to select a subsample of PIKs. This is extracted from the first two digits of the PIK, approximately uniformly distributed on [00, 99]. Note that "AA" is also a valid value, denoting individuals for whom no valid SSN was on file. Occurence of such "pseudo-PIKs" varies by state.

2.1.3 Codebook: The EHF Files

Table Metadata for Employment History File (EHF_ZZ)

Table 1: Access Requirements for EHF_ZZ

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description
Quarterly earnings for each job in a state as reported by the employer to the state's UI system.

Scope
State

Key
PIK SEIN SEINUNIT YEAR

Sort Order
PIK SEIN SEINUNIT YEAR

Variable Information

Table 2: Variable Information for EHF_ZZ

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
YEAR	num	4	Calendar year
EARN_ANN	num	8	Annual earnings
EARN1	num	5	Qtr 1 earnings
EARN2	num	5	Qtr 2 earnings
EARN3	num	5	Qtr 3 earnings
EARN4	num	5	Qtr 4 earnings
SEIN	char	12	State Employer Identification Number
SEINUNIT	char	5	State UI reporting unit
STATE	char	2	Geographic state of the job record. For UI, it is the FIPS code of the state being processed, for OPM, it is the duty state as reported on the OPMUI files
FLAG_IMPUTE1	char	1	Qtr 1 imputation flag
FLAG_IMPUTE2	char	1	Qtr 2 imputation flag
FLAG_IMPUTE3	char	1	Qtr 3 imputation flag
FLAG_IMPUTE4	char	1	Qtr 4 imputation flag

2.1.4 Codebook: The JHF Files

Table Metadata for Job History File (JHF_ZZ)

Table 3: Access Requirements for JHF_ZZ

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description
Wide version of earnings history that contains imputed establishment (SEINUNITs) for multi-unit firms, and a job-level ID useful for tracking job spells across employer identifier changes.

Scope
State

Key
PIK SEIN SPELL_U2W

Sort Order
PIK FID SEIN SPELL_U2W

Variable Information

Table 4: Variable Information for JHF_ZZ

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
SEIN	char	12	State Employer Identification Number
SPELL_U2W	num	3	
E21-E147	num	8	Earnings (*See details below*)
FID	num	5	Within-PIK linked job spell identifier
SEINUNIT1	char	5	State UI Reporting Unit Number (Impute 1)
SEINUNIT2	char	5	State UI Reporting Unit Number (Impute 2)
SEINUNIT3	char	5	State UI Reporting Unit Number (Impute 3)
SEINUNIT4	char	5	State UI Reporting Unit Number (Impute 4)
SEINUNIT5	char	5	State UI Reporting Unit Number (Impute 5)
SEINUNIT6	char	5	State UI Reporting Unit Number (Impute 6)
SEINUNIT7	char	5	State UI Reporting Unit Number (Impute 7)
SEINUNIT8	char	5	State UI Reporting Unit Number (Impute 8)
SEINUNIT9	char	5	State UI Reporting Unit Number (Impute 9)
SEINUNIT10	char	5	State UI Reporting Unit Number (Impute 10)
FIRST_ACC	num	3	First accession (First quarter of employment at this job spell)
LAST_SEP	num	3	Last separation (Last quarter of employment at this job spell)
FLAG_SEINUNIT_IMPUTED	num	3	
STATE	char	2	
RANDOM_PIK_GROUP	char	2	Selector based on random pik

Details for variables E21-E147 on JHF_ZZ

Back

Description
Earnings

Source
UI Records

Valid Values
Positive and negative integer values.

Notes

The set of quarterly variables available in each state corresponds to the set of quarters for which earnings data are available for that state.

2.1.5 Codebook: The EHF_US_INDICATORS File

Table Metadata for Earnings Indicators (EHF_US_INDICATORS)

Table 5: Access Requirements for EHF_US_INDICATORS

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			

Description

Indicates whether or not a PIK had any earnings reported in any state in the LEHD system.

Scope

National

Key

PIK YEAR

Variable Information

Table 6: Variable Information for EHF_US_INDICATORS

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
YEAR	num	3	Calendar year
NUM_STATES_EARN1	num	3	Number of states reporting earnings for PIK in Q1
NUM_STATES_EARN2	num	3	Number of states reporting earnings for PIK in Q2
NUM_STATES_EARN3	num	3	Number of states reporting earnings for PIK in Q3
NUM_STATES_EARN4	num	3	Number of states reporting earnings for PIK in Q4

2.1.6 Codebook: The EHF_ALL_AVAILABILITY File

Table Metadata for Earnings Availability (EHF_ALL_AVAILABILITY)

Table 7: Access Requirements for EHF_ALL_AVAILABILITY

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			

Description

Indicates data availability ranges for all states.

Scope

National

Key
State

Variable Information

Table 8: Variable Information for EHF_ALL_AVAILABILITY

Variable Name	Type	Length	Description
STATE	char	2	State postal code
START_YEAR	num	8	First year of data on EHF
START_QUARTER	num	8	First quarter of data on EHF
END_YEAR	num	8	Last year of data on EHF
END_QUARTER	num	8	Last quarter of data on EHF
START_YEAR_JHF	num	8	First year of data on JHF
START_QUARTER_JHF	num	8	First quarter of data on JHF
END_YEAR_JHF	num	8	Last year of data on JHF
END_QUARTER_JHF	num	8	Last quarter of data on JHF
START_QTIME_JHF	num	3	Last quarter of data on JHF (1985Q1=1)
END_QTIME_JHF	num	3	Last quarter of data on JHF (1985Q1=1)

CHAPTER

THREE

EMPLOYER-LEVEL FILES

3.1 Employer Characteristics File (ECF)

3.1.1 Overview

This file family contains industry, employment size, geography, and ownership of employers. The primary source of characteristics is the Quarterly Census of Employment and Wages (QCEW). Establishment addresses have been geocoded to assign geographic regions. National firm age and size information are largely sourced from the Census Bureau Longitudinal Business Database (these require IRS approval).

The ECF file family includes the following files:

- **ECF_ST_SEINUNIT**

  An establishment level file with industry, size, location, and ownership.

  - Scope: State
  - Key: SEIN YEAR QUARTER SEINUNIT

- **ECF_ST_SEIN**

  Sourced from the SEINUNIT file, this file provides SEIN-level modal characteristics across establishments. Reference employment counts from UI wage data are also included, as available.

  - Scope: State
  - Key: SEIN YEAR QUARTER

- **ECF_ST_T26:**

  This table contains the national firm age and size for employers in the ECF. This file requires IRS approval.

  - Scope: State
  - Key: SEIN YEAR QUARTER

3.1.2 User Guidance

Identifiers: Employers vs. Firms vs. Establishments

QCEW data are reported at the establishment (SEINUNIT) level, regardless of state. However, UI wage records in nearly all states are reported at the state tax identifier level (SEIN), and users often wish to conduct analyses at an employer or firm level. We typically refer to SEINs as *employers* in this document, but users should be aware they are fundamentally a tax entity and distinct from an establishment or a firm, as described below:

- For single-unit firms the SEIN is the firm and the establishment. These employ about half of workers.
- For multi-unit firms the SEIN is sometimes a firm and sometimes a subset of firm activity. Firms will have multiple SEINs if the firm has employment across multiple states, but can also have mutiple SEINs within a state.
- Multi-unit SEINs indicate the firm has multiple establishments within a state. It is most common that establishments within a SEIN operate in the same industry, but there are exceptions.
- Information on the parent firm is available via links to IRS information on firm ownership, obtained from the Longitudinal Business Database (LBD) [Chow *et al.*, 2021] . These parent firm identifiers, as well as parent firm age and size characteristics, may be obtained by linking to the Title 26 ECF file.

The ECF SEINUNIT file contains establishment-level information that is primarily provided directly from QCEW. In contrast, the ECF SEIN file contains information about the modal characteristics at the tax identifier (SEIN) level, which is constructed from the establishment-level records. If your project makes use of the LEHD's job-level files, then you are able to link these ECF files directly to the EHF, JHF, or QWI to incorporate establishment or employer-level information. However, depending on your use case, you may wish to construct additional linkages at a level that corresponds with the LBD definition of a firm. Any such linkages require the use of the ECF T26 file, and will therefore require IRS approval.

For further details on the firm-level age and size variable construction, *see below*.

Establishment Geocoding

Geography is provided on the QCEW inputs in several different fields:

- Physical address of establishment
- Mailing address for establishment
- County of establishment

The physical address is the preferred source to assign geographic information in the ECF, followed by the mailing address. Both of these addresses are geocoded in the GAL process (Geocoded Address List) to assign a Census block from the most recent Decennial tabulation. The geocoding of each address is given a quality code, 1-4 for physical and 5-8 for mailing. These can be thought of as general reductions in precision, where a code of 1 or 5 may mean that the address was placed in a Census block, and 4 or 8 may mean that only the zip code could be used, so the geocodeed block is not considered a precise location assignment. Mailing addresses are considered inferior to the physical address, since they are not always the actual place of work. When both of these assignments fail or data are missing, a fixed block is selected for the reported county with a geographic quality flag of 9.

After the initial assignment of the Census block, a series of longitudinal edits is performed to improve consistency over time, typically using higher quality information in preference to lower quality information in other quarters. Establishments are allowed to "move," but it can be difficult to determine whether those moves are a true relocation or simply an artifact of changes in addressing practices. Note, unusually high rates of flux in geographic assignments have been noted in certain state-quarters.

If no establishment geography is available after these edits, a county is imputed for the establishment conditional on industry, using observed statewide data for a posterior. These records are given a geo quality flag of 10.

Tabulation vs. Current Geography

The GEOID_YYYY variable is a 15-character key that references a Census block defined in the latest decennial Census. This is referred to as the tabulation geography. However, regional boundaries and defintions evolve over time, so the codes baked into the GEOID are not necessarily the latest available. The ECF provides a set of regions in *current* geography that are valid at the time the ECF is produced.

For multi-establishment employers, employment-weighted modal regions are provided at the SEIN level. These regions are not necessarily consistent with each other. For example, though CBSAs (Core-Based Statistical Areas) are composed of counties, the modal county for a firm may not be a component of the modal CBSA, etc.

Industry Assignment

Industry is reported at the establishment level using SIC codes in earlier years of data and various iterations of NAICS codes up to the present. The reported codes are run through a transition algorithm to provide industries in all other coding systems, SIC (1987) and NAICS (1997-2017). The basic steps of the algorithm are as follows:

1. Identify the most likely coding system used to assign the industry to establishments. This is done with a comparison of reported codes to valid codes in each reporting system.
2. Retrieve industry tables provided at https://www.census.gov/naics/ to establish concordances between industry coding systems.
3. Use observed coding changes on establishment data in consecutive quarters to calculate the shares of establishments that get recoded based on the expected concordances. This forms a posterior for imputation.
4. Perform a combination of longitudinal edits and imputation to generate an industry assignment in all coding systems.
    - If an industry before and after a coding system transition is consistent with a pairing on the concordance tables, longitudinal edits are applied to maintain industry codes.
    - Other industry code changes are considered a change in the primary activity of the establishment. Longitudinal edits are not applied across these changes.
    - Industry codes are cast into each earlier and subsequent taxonomy as needed by using the posterior calculated from observed transition.
5. Imputation using a statewide posterior is employed to provide an industry to an establishment for which none is reported.

Only the final industry codes assigned in each system are kept on the ECF. The SEIN level table provides the employment-weighted modal employment calculated across establishments.

Ownership

Ownership is provided on the establishment level, but it is extremely unusual for two establishments in the same SEIN to report different ownership codes. Any variations in this pattern could indicate an anomaly in the data. Employers with federal ownership are included in the ECF, but these are not associated with wage records (except in extremely isolated and limited instances).

Firm Age and Size

Firm age and size characteristics are assigned based on the national firm. Firms are defined at the level of operational control (alpha), and so a single firm can include multiple federal EINs. Firm age and size are defined in terms of employment.

Firm age is defined as the age of the oldest establishment when a new alpha comes into existence. Age measures the number of years from the first year of employment (i.e., the first year with positive payroll), when the age is zero. New establishments of an older firm are assigned the age of the national firm.

National firm size is the sum of all workers employed at each establishment in the national (alpha) firm on March 12th of the year. Firm size in year t is defined as the employment-weighted sum of firm size on March 12 in year t-1 of all establishments that are part of an EIN on March 12 in year t.

The primary source of firm age and size information is the Longitudinal Business Database (LBD), an annual table derived from federal tax information (FTI). Because firm age and size data are sourced from FTI, these employer characteristics are included in the ECF T26 files. Employers (SEINs) are linked to the LBD via the federal EIN. When a link to the LBD cannot be made, the age and size is developed from ECF data using a combination of edits and imputation. For additional methodological details, see Haltiwanger *et al.* [2014].

Federal EIN and FAS_EIN Variable

The FAS_EIN variable is attached to the SEIN to assign firm age and size characteristics. When the Federal EIN is available on input data, it is incorporated into the FAS_EIN, though in some cases it may be encrypted and not linkable to the EIN. The EIN may be edited from that originally reported for longitudinal consistency or to improve linkages with the LBD. When the EIN is not available, the FAS_EIN is formed from the SEIN, with two digits added for a counter to handle cases where the SEIN appears to have been recirculated by the state (unrelated firms being assigned the same SEIN). The FAS_EIN may appear as follows:

- Prefix **EINUS** - last 9 digits are the Federal EIN (e.g., `EINUS123456789`)
- Prefix **EIN[state]** - last 9 digits are an encrypted Federal EIN (e.g., `EINNY123456ABC`)
- Otherwise, SEIN plus increment (e.g., `33ABCD1234EFGH01`)

3.1.3 Codebook: The ECF SEINUNIT Files

Table Metadata for Employer Characteristics File, Establishment (ECF_ZZ_SEINUNIT)

Table 1: Access Requirements for ECF_ZZ_SEINUNIT

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description
An establishment level file with industry, size, and location.

Scope
State

Key
SEIN YEAR QUARTER SEINUNIT

Sort Order
SEIN YEAR QUARTER SEINUNIT

Variable Information

Table 2: Variable Information for ECF_ZZ_SEINUNIT

Variable Name	Type	Length	Description
ES_STATE	char	2	ES202 FIPS State SS
SEIN	char	12	State Employer Identification Number
SEINUNIT	char	5	State establishment identifier
YEAR	num	4	Year (YYYY)
QUARTER	num	4	Quarter (Numeric)
QTIME	num	3	Quarter index (1985Q1=1)
IN_202	num	3	SEIN in ES202
IN_UI	num	3	SEIN in UI
EMPL_MONTH1	num	8	ES202 Employment Month 1, edited
EMPL_MONTH1_FLG	char	1	Reported or imputed Month 1 Employment
EMPL_MONTH2	num	5	ES202 Employment Month 2, edited
EMPL_MONTH2_FLG	char	1	Reported or imputed Month 2 Employment
EMPL_MONTH3	num	5	ES202 Employment Month 3, edited
EMPL_MONTH3_FLG	char	1	Reported or imputed Month 3 Employment
TOTAL_WAGES	num	7	ES202 wages, edited
TOTAL_WAGES_FLG	char	1	Reported or imputed Total Wages
BEST_EMP1	num	8	Best UI/202 Employment Month 1
BEST_EMP2	num	8	Best UI/202 Employment Month 2
BEST_EMP3	num	8	Best UI/202 Employment Month 3
BEST_WAGES	num	8	Best UI/202 Wages
ES_OWNER_CODE	char	1	Owner code, edited
GEOID_2020	char	15	Census geocode: Tabulation state FIPS (2) || Tabulation county FIPS (3) || tract (6) || block (4)
LEG_STATE	char	2	Current FIPS State
LEG_COUNTY	char	5	
LEG_LATITUDE	num	8	Internal point latitude, 6 implied decimals
LEG_LONGITUDE	num	8	Internal point longitude, 6 implied decimals
LEG_GEO_QUAL	num	3	Quality of final geography
LEG_TRACT	char	6	Current Census Tract
LEG_BLKGRP	char	1	Current Block Group
LEG_BLOCK	char	4	Census Block
LEG_CBSA	char	5	Core-Based Statistical Area
LEG_BLOCK_SUF1	char	1	Census Block suffix 1
LEG_BLOCK_SUF2	char	1	Census Block suffix 2
LEG_CBSA_MEMI	char	1	CBSA type 1=Metro, 2=Micro, else=9
LEG_WIB	char	8	
LEG_SSCCC	char	5	
SIC1987FNL	char	6	Final SIC 1987 code
NAICS1997FNL	char	6	Final NAICS 1997 code
NAICS2002FNL	char	6	Final NAICS 2002 code
NAICS2007FNL	char	6	Final NAICS 2007 code
NAICS2012FNL	char	6	Final NAICS 2012 code
NAICS2017FNL	char	6	Final NAICS 2017 code
RAMP_SEIN	num	8	Ramp Fuzz Factor
UNIFORM_SEIN	num	8	Draw from Uni Dist
RAMP_Z_SEIN	num	8	Ramp Fuzz Factor

continues on next page

Table 2 – continued from previous page

Variable Name	Type	Length	Description
UNIFORM_Z_SEIN	num	8	Draw from Uni Dist

3.1.4 Codebook: The ECF SEIN Files

Table Metadata for Employer Characteristics File, Firm (ECF_ZZ_SEIN)

Table 3: Access Requirements for ECF_ZZ_SEIN

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description
Sourced from the SEINUNIT file, this file rolls up employer characteristics to the SEIN level.

Scope
State

Key
SEIN YEAR QUARTER

Sort Order
SEIN YEAR QUARTER

Variable Information

Table 4: Variable Information for ECF_ZZ_SEIN

Variable Name	Type	Length	Description
ES_STATE	char	2	ES202 FIPS State SS
SEIN	char	12	State Employer Identification Number
YEAR	num	4	Year (YYYY)
QUARTER	num	4	Quarter (Numeric)
QTIME	num	3	Quarter index (1985Q1=1)
IN_202	num	3	SEIN in ES202
IN_UI	num	3	SEIN in UI
SEIN_EMP1	num	8	SEIN 202 Employment Month 1
SEIN_EMP2	num	8	SEIN 202 Employment Month 2
SEIN_EMP3	num	8	SEIN 202 Employment Month 3
SEIN_WAGES	num	8	SEIN 202 Wages
SEINSIZE_M	num	8	UI Employment M
SEINSIZE_B	num	8	UI Employment B
SEINSIZE_E	num	8	UI Employment E
PAYROLL	num	8	Original UI Payroll Info W1
SEIN_BEST_EMP1	num	8	SEIN Best UI/202 Month 1, Employment
SEIN_BEST_EMP2	num	8	SEIN Best UI/202 Month 2, Employment
SEIN_BEST_EMP3	num	8	SEIN Best UI/202 Month 3, Employment
SEIN_BEST_WAGES	num	8	SEIN Best UI/202 Payroll
BEST_FLAG	num	8	Source of best_ data
MULTI_UNIT	num	3	SEIN with 2 or more establishments on 202
NUM_ESTABS	num	4	Number of Establishments

continues on next page

Table 4 – continued from previous page

Variable Name	Type	Length	Description
MODE_ES_OWNER_CODE_EMP	char	1	Employment mode owner code
MODE_ES_COUNTY_EMP	char	5	
MODE_LEG_STATE_EMP	char	2	
MODE_LEG_COUNTY_EMP	char	5	
MODE_LEG_WIB_EMP	char	8	
MODE_LEG_CBSA_EMP	char	5	Employment mode Core-based statistical area
MODE_SIC1987FNL_EMP	char	6	Employment mode Final SIC 1987 code
MODE_NAICS1997FNL_EMP	char	6	Employment mode Final NAICS 1997 code
MODE_NAICS2002FNL_EMP	char	6	Employment mode Final NAICS 2002 code
MODE_NAICS2007FNL_EMP	char	6	Employment mode Final NAICS 2007 code
MODE_NAICS2012FNL_EMP	char	6	Employment mode Final NAICS 2012 code
MODE_NAICS2017FNL_EMP	char	6	Employment mode Final NAICS 2017 code
RAMP_SEIN	num	8	Ramp Fuzz Factor
UNIFORM_SEIN	num	8	Draw from Uni Dist
RAMP_Z_SEIN	num	8	Ramp Fuzz Factor
UNIFORM_Z_SEIN	num	8	Draw from Uni Dist

3.1.5 Codebook: The ECF T26 Files

Table Metadata for Employer Characteristics File, National Firm (ECF_ZZ_SEIN_T26)

Table 5: Access Requirements for ECF_ZZ_SEIN_T26

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X	X	

Description
Contains the national firm age and size data for employers in the ECF. This file requires IRS approval.

Scope
State

Key
SEIN YEAR QUARTER

Sort Order
SEIN YEAR QUARTER

Variable Information

Table 6: Variable Information for ECF_ZZ_SEIN_T26

Variable Name	Type	Length	Description
SEIN	char	12	State Employer Identification Number
YEAR	num	4	Year (YYYY)
QUARTER	num	4	Quarter (Numeric)
FIRMID	char	10	Firm Alpha from LBD
FAS_EIN	char	14	Firm identifier for Age & Size
FAS_EIN_FLAG	num	3	Info about fas_ein variable
FIRMAGE	num	8	Firm Age, calculated
SOURCE_AGE	char	1	Source of firm age
FIRMSIZE	num	8	Best initial firm size (size March 12 of last year, current size if new)
SOURCE_SIZE	char	1	Source of firm size
FIRM_BIRTH_YEAR	num	8	Completed year of birth
FIRM_BIRTH_QUARTER	num	8	Completed quarter of birth
FIRMSIZE_FUZZ	num	8	Noise infused value of firmsize
LBD_MATCH	num	8	1=EIN matches to LBD this year,2=next year,3=previous year,4=NECF(no LBD),5=ECF only
MULTI_UNIT_LBD	num	8	Multi Unit firm in LBD microdata

3.2 Successor-Predecessor File (SPF)

3.2.1 Overview

The firm identifier used in the LEHD infrastructure tables is a state-specific unemployment insurance (UI) account number used to administer the tax and benefits of the UI system. These account numbers, prepended by a state identifier and here called "SEIN," can change for a number of reasons, including a simple change in legal form, a merger with another firm, or a divestiture into several firms. Typically, the separation of a worker from an employer is identified by a change in the SEIN on that workers' UI wage records. If an employer changes SEINs but makes no other changes, the worker would appear to have left the original firm even though his or her employment status remains unchanged from an economic viewpoint. These spurious employer changes are known to induce biases in both employment and job flow statistics. For example, a simple change in account numbers would lead to the observation of a firm closing and all workers separating, even though all (or most) workers remain employed.

- **Successor-Predecessor File (SPF_ZZ)**
  - Scope: State
  - Key: SEIN SEIN_SUCC QTIME

To identify and correct for such events, the Successor-Predecessor File (SPF) was created to track large worker movements between SEINs using quarterly UI wage records. Benedetto *et al.* [2007] used the SPF for an early analysis in one particular state of the impact of such an exercise. The SPF provides link characteristics based on the number of workers leaving an origin SEIN and entering a destination SEIN, presented in absolute terms and relative terms to the overall size of the firms. In addition to the UI-sourced information, predecessor and/or successor firms may be reported directly on the firm microdata from the Quarterly Census for Employment and Wages (QCEW or ES202), and these linkages are also recorded in the SPF. Note, links are identified using UI data for one state at a time. Cross-state firm relocations are not identified here, and such events would be interpreted as separations in the origin state and accessions in the destination state.

Research use of the SPF

The linkages identified in the SPF have already been incorporated into jobs on the JHF and QWI tables as described in *Accounting for Successor-Predecessor Events*. The JHF is the recommended input for analyses of job starts, tenure, separations, at the job level, and the QWI can be used for establishment-level calculations of job flows. The SPF itself may not be needed by most researchers.

Researchers who wish to explore an alternate method of accounting for SEIN changes (e.g., modifying the thresholds that identify firm linkages) may want to use the full set of flows presented in the SPF. Researchers interested in the evolution of firm structure across mergers and divestures may also benefit from linkages found in the SPF.

3.2.2 User Guidance

Identifying Transitions in UI Data

SPF processing starts by finding all job-level separations and accessions from the quarterly UI data. A separation in period t is defined for PIK *i* from firm *j* when *i* receives earnings from *j* in quarter *t*, but not in *t+1*. Similarly, an accession in period *t* is defined for PIK *i* into firm *k* when *i* receives earnings from *k* in quarter *t* but did not in *t-1*. Then for every PIK, separations in period *t* are linked with all accessions in periods *t* or *t+1* to identify possible candidates for SEIN transitions at the job level. The timing in this linkage strategy recognizes that a transition could be reported to occur within a quarter or across quarters, or indeed a combination of both at the firm level.

These SEIN transitions found at the PIK level are then aggregated across PIKs to get the count of transitions from predecessor *j* (SEIN) to successor *k* (SEIN_SUCC) in period *t* (*qtime*), where *t* is the quarter of separation from *i* (accession to *j* in *t* or *t+1*). The count of PIKs making the transition is recorded in the NUM_LEFT variable. For informational purposes, the MATCH_PERIOD variable is created to report the share of jobs for which the accession occurred in *t+1*. The *de minimis* SEIN transitions are ignored by removing links where the NUM_LEFT is less than 6. The remaining records form the UI-sourced frame of the SPF.

Assessing Strength of SEIN Transitions in UI Data

Several factors are used to assess whether a linkage between two firms in a quarter constitutes a predecessor-successor relationship:

- The share of employment at the predecessor that transits to the successor
- The share of employment at the successor that transited from predecessor
- Death of the predecessor firm, such that UI records are no longer reported after the transition
- Birth of the successor firm, such that UI records were not reported prior to the transition

The shares of jobs are provided in the variables RATIO (predecessor) and SUCC_RATIO (successor). Because of timing issues employment patterns and transitions, the numerator (the count of jobs in the transition) is not always a strict subset of the employment count at the firm, so the ratios are considered estimates of the share of the firm

employment that is part of the transition. The ratio is capped at 100%. A ratio of 80% is considered the threshold for representing a large share of the predecessor or successor.

Composite flags LINK_UI and SUCC_LINK_UI are then created to combine the strength of the link with information on firm birth and death. The values of the flags are as follows (description for SUCC_LINK_UI in parentheses):

Table 7: LINK_UI (SUCC_LINK_UI) Flag Values

1	Firm death (birth), flow constitutes a large percentage of predecessor (successor) employment
2	Firm death (birth), flow does not constitute a large percentage of predecessor (successor) employment.
4	Not firm death (birth), flow constitutes a large percentage of predecessor (successor) employment
5	Not firm death (birth), flow does not constitute a large percentage of predecessor (successor) employment.

Directly Reported Links from Firm Data

The UI sourced flows on the SPF are supplemented with links that are provided on the firm microdata (ES202). In each quarter, the firm can be reported with having a predecessor, successor, or both. These links are collected and merged onto the UI-sourced links by quarter. The SOURCE flag is created to identify which source provided the SEIN transition in a quarter, with values as follows:

Table 8: SOURCE Flag Values

UI	Wage record flow observed in UI data, only
ES	Transition reported in ES202 data, only
UIES	Wage record flow observed in UI and reported in ES202 data

Because the timing of the report on the ES202 and the observation of the event in UI data do not always coincide precisely, an ES202 transition reported between two firms in a proximate quarter will be flagged as happening at the time the UI transition is observed.

Identifying Successor-Predecessor Relationships

For LEHD data products, only the strongest links based on the UI wage record flows are used to filter out spurious employer identifier changes. A link between two SEINs is also used if it was directly reported on the input ES202 data, even if it does not meet the UI requirements. The following are the precise business rules used:

- 80% of jobs at the predecessor are linked to accessions at the successor, and the predecessor firm dies after the transition.
  - LINK_UI=1
- 80% of jobs at the successor are linked to separations from the predecessor, and the successor firm is born in the transition.
  - SUCC_LINK_UI=1
- The separations in the flow represent 80% of employment at the successor, and the accessions in the flow represent 80% of employment at the successor.
  - LINK_UI=4 AND SUCC_LINK_UI=4
- A flow of wage records is identified between firms in a quarter, and the predecessor and/or the successor SEIN is reported on the ES202.
  - SOURCE="UIES"

Successor-predecessor relationships that meet at least one of these requirements are used to link job spells in the Job History File (JHF). The process of job linkage is further described in the documentation for the JHF.

3.2.3 Codebook

Table Metadata for Successor-Predecessor File (SPF_ZZ)

Table 9: Access Requirements for SPF_ZZ

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description

Flows of separations and hires between employers (SEINs) used to identify successor-predecessor relationships.

Scope

State

Key

SEIN SEIN_SUCC QTIME

Sort Order

SEIN SEIN_SUCC QTIME

Variable Information

Table 10: Variable Information for SPF_ZZ

Variable Name	Type	Length	Description
SEIN	char	12	SEIN - predecessor
QTIME	num	8	Quarter of separation, 1985Q1=1
SEIN_SUCC	char	12	SEIN - successor
MATCH_PERIOD	num	8	Percent of transitions where separation precedes quarter of accession
NUM_LEFT	num	8	Number of jobs transitioning between firms
ACTIVE_BEG_QTR_A	num	3	First quarter predecessor is active on UI
ACTIVE_END_QTR_A	num	3	Last quarter predecessor is active on UI
ACTIVE_BEG_QTR_B	num	3	First quarter successor is active on UI
ACTIVE_END_QTR_B	num	3	Last quarter successor is active on UI
LINK_UI	num	3	Type of link for predecessor firm
SUCC_LINK_UI	num	3	Type of link for successor firm
BPEMP_MASTER	num	8	Pred UI B Employment, max of last 3 quarters
RATIO	num	8	Percent of jobs at predecessor transitioning to successor (estimated)
SUCC_RATIO	num	8	Percent of jobs at successor transitioning from predecessor (estimated)
PRED_SIZE_CLASS_UI	num	8	Size class of predecessor based on UI data (B employment in qtime)
SUCC_SIZE_CLASS_UI	num	8	Size class of successor based on UI data (M employment in qtime+1)
EMP_ES	num	8	Pred ES202 Month 1 Employment, max of last three quarters
ES_QTRS_OFF	num	3	Number of quarters removed ES202 event date is from UI flow
SOURCE	char	4	Data source of link between firms

3.3 Quarterly Workforce Indicators (QWI)

3.3.1 Overview

The Quarterly Workforce Indicators (QWI) are a set of measures of workforce composition and worker turnover calculated from unemployment insurance wage records. The QWI SEINUNIT table contains these measures calculated at the establishment level. Internally, this table is referenced as the Unit Flows File (UFF_B).

- **Quarterly Workforce Indicators (QWI_ZZ_SEINUNIT)**
  - Scope: State
  - Key: SEIN YEAR QUARTER SEINUNIT

Establishment-level measures include:

- Worker and Job Flows: accessions, separations, job creation, job destruction

- Worker compensation for stocks and flows
- Dynamic worker compensation summary statistics for stocks and flows

The QWI may be used in combination with the ECF to match to other Census micro business databases and can be linked by employer-establishment identifiers to other LEHD Infrastructure files. Aggregated versions of some of these measures are available in the QWI public use data.

3.3.2 User Guidance

QWI Calculation

QWI measures are calculated at the job level using the longitudinal work history of the worker at the employer. The work history has been corrected to account for successor-predecessor relationships (jobs reported to multiple SEINs which should be considered a single spell of employment). Measures are weighted to account for multiple imputation as well as successor-predecessor reporting issues. Measures are then aggregated to the establishment level for the QWI microdata table. Job creation, job destruction, and a number of rates are calculated after aggregation.

Because only quarterly earnings are available in the job history, patterns of zero or positive earnings are referenced to generate QWI count measures. Some basic concepts used in defining QWI measures follows. The full definitions of QWI measures are presented in Abowd *et al.* [2009].

- **Flow employment**
  Any job with positive earnings in quarter *t* is counted towards flow employment (**M**) in *t*. It is noted that this measure may contain a significant fraction of short duration jobs that do not survive the quarter.
- **Point-in-time employment**
  A job with positive earnings in quarters *t* and *t+1* is inferred to be active at the seam between the quarters. This counts as an end-of-quarter job (**E**) in *t*, and a beginning-of-quarter job (**B**) in *t+1*.
- **Separations**
  A job with earnings in *t* and no earnings in *t+1* is counted as a separation (**S**) in *t*. Separations from continuning quarter employment (**CS**) is a recommended measure, as it ignores noise from short duration jobs.
- **Accessions**
  A job with no earnings in *t-1* and earnings in the *t* is counted as an accession (**A**) in *t*. Accessions into continuning quarter employment (**CA**) is a recommended measure, as it ignores noise from short duration jobs.
- **Full quarter**
  A job with earnings in *t-1*, *t*, and *t+1* is considered full quarter employed (stable) in *t*. This concept is used to calculate measures of average earnings per job, removing jobs that are active for only part of the quarter. Note, this could represent a mix of full- and part-time jobs.

The number of leading or lagging quarters of data required for the calculation of a measure is presented on the detailed tables that follow.

QWI Count Measures

These primary counts are calculated at the job level and aggregated to the establishment level. A job-level reference is provided to assist in cross-referencing variables in Abowd *et al.* [2009]. For corresponding measures that are released in QWI public use files, the public use name is given.

Table 11: QWI Count Measures

QWI Variable	Job Reference	Description	Leading Quarters	Lagging Quarters	Public Use QWI
B	b	Beginning-of-period employment	1	0	Emp
E	e	End-of-period employment	0	1	EmpEnd
M	m	Employment any time during the period	0	0	EmpTotal
F	f	Full-quarter employment	1	1	EmpS
Fpv		Full-quarter employment - previous quarter	2	1	EmpSpv
A	$a1$	Accessions	1	0	HirA
CA	$a2$	Accession into Continuing Quarter Employment	1	1	HirAEnd
FA	$a3$	Flow into full-quarter employment	2	1	HirAS
S	$s1$	Separations	0	1	Sep
CS	$s2$	Separations from Continuing Quarter Employment	1	1	SepBeg
FS	$s3$	Flow out of full-quarter employment	2	1	SepS
$FSnx$		Flow out of full-quarter employment - next quarter	1	2	SepSnx
H	$h1$	New hires	4	0	HirN
CH	$h2$	New Hires into Continuous Quarter Employment	4	1	
$H3$	$h3$	Full-quarter new hires	5	1	HirNS
R	$r1$	Recalls	4	0	HirR
CR	$r2$	Recalls into Continuous Quarter Employment	4	1	
$W1$	$w1$	Total payroll of all employees	0	0	Payroll

QWI Earnings

Total earnings for select full quarter job histories are aggregated to the establishment level. These are presented as average monthly earnings, dividing the quarterly earnings by 3. The base count associated with the total earnings is given. A job-level reference is provided to assist in cross-referencing variables in Abowd *et al.* [2009]. For corresponding measures of earnings *per job* released in QWI public use files, the public use name is given.

Table 12: QWI Earnings

QWI Variable	Job Reference	Description	Leading Quarters	Lagging Quarters	Earnings Base	Public Use QWI
$W2B$	$w2b$	Average monthly earnings of beginning-of-period employees	1	0	B	EarnBeg
$W3$	$w3$	Average monthly earnings of full-quarter employees	1	1	F	EarnS
WFA	$wa3$	Average monthly earnings of transits to full-quarter status	2	1	FA	EarnHirAS
$WH3$	$wh3$	Average monthly earnings of new hires to full-quarter status	5	1	$H3$	EarnHirNS
WFS	$ws3$	Average monthly earnings of separations from full-quarter status	1	2	$FSnx$	EarnSepS

Establishment-level Calculations

After aggregation to the establishment level, a number of measures are created based on the primary counts. For corresponding measures that are released in QWI public use files, the public use name is given.

Table 13: QWI Establishment-level Calculations

QWI Variable	Description	Leading Quarters	Lagging Quarters	Calculation	Public Use QWI
$Ebar$	Average employment	1	1	$(E+B)/2$	
JF	Net job flows	1	1	$E-B$	FrmJbC
JC	Job creation	1	1	$max(0, JF)$	FrmJbGn
JCR	Average job creation rate	1	1	$JC/Ebar$	
JD	Job destruction	1	1	$\|min(0, JF)\|$	FrmJbLs
JDR	Average job destruction rate	1	1	$JD/Ebar$	
$Fbar$	Average full-quarter employment	2	1	$(F+Fpv)/2$	
FJF	Net change in full-quarter employment	2	1	$F-Fpv$	FrmJbCS
FG	Average full-quarter employment growth rate	2	1	$FJF/Fbar$	
FJC	Full-quarter job creation	2	1	$max(0, FJF)$	FrmJbGnS
$FJCR$	Average full-quarter job creation rate	2	1	$FJC/Fbar$	
FJD	Full-quarter job destruction	2	1	$\|min(0, FJF)\|$	FrmJbLsS
$FJDR$	Average full-quarter job destruction rate	2	1	$FJD/Fbar$	
AR	Average accession rate	1	1	$A/Ebar$	
SR	Average separation rate	1	1	$S/Ebar$	
FAR	Average rate of flow into full-quarter employment	2	1	$FA/Fbar$	
FSR	Average rate of flow out of full-quarter employment	2	1	$FS/Fbar$	

Establishment Characteristics on QWI Table

The establishment characteristics on the QWI table are acquired through a lookup to the *Employer Characteristics File (ECF)*, using a key of SEIN-SEINUNIT-YEAR-QUARTER. Because the establishment may not exist on the ECF in all quarters it appears on the QWI table, a longitudinal lookup may be used to get characteristics from a following quarter. This is indicated in the UNIT_QUARTERS_OFF variable. For more information on this lookup, see *Adding Establishment Characteristics*.

3.3.3 Codebook

Table Metadata for Quarterly Workforce Indicators (QWI_ZZ_SEINUNIT)

Table 14: Access Requirements for QWI_ZZ_SEINUNIT

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements	X		

Description
The microdata version of the public use QWI data, containing establishment-level hires, separations, net job growth, and average earnings.

Scope
State

Key
SEIN YEAR QUARTER SEINUNIT

Sort Order
SEIN YEAR QUARTER SEINUNIT

Variable Information

Table 15: Variable Information for QWI_ZZ_SEINUNIT

Variable Name	Type	Length	Description
SEIN	char	12	State Employer ID Number
SEINUNIT	char	5	State UI Reporting Unit Number
YEAR	num	4	Year YYYY
QUARTER	num	4	Quarter QQ
ES_OWNER_CODE	char	1	Cleaned OWNER_CODE O
GEOID_2020	char	15	Census geocode: Tabulation state FIPS (2) || Tabulation county FIPS (3) || tract (6) || block (4)
LEG_GEO_QUAL	num	3	Quality of final geography
SIC1987FNL	char	6	Cleaned SIC Code IIII
NAICS2017FNL	char	6	Final 2017 NAICS Code NNNNNN
DELTA	num	8	Ramp fuzz factor
DELTAZ	num	8	Ramp fuzz factor- Z measures
LEG_CBSA	char	5	Core-Based Statistical Area
LEG_STATE	char	2	ES202 FIPS State SS
LEG_COUNTY	char	5	
LEG_WIB	char	8	
QWI_UNIT_WEIGHT	num	3	UI this quarter and ES in at least one quarter

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Table 15 – continued from previous page

Variable Name	Type	Length	Description
ES_STATE	char	2	
UNIT_QUARTERS_OFF	num	3	Number of quarters away from data that establishment was found
QWI_FINAL_WEIGHT	num	3	In-scope for QWI tabulation
B	num	8	Beginning-of-period employment
E	num	8	End-of-period employment
M	num	8	Employment any time during the period
F	num	8	Full-quarter employment
H3	num	8	Full-quarter new hires
W1	num	8	Total payroll of all employees
W2B	num	8	Average monthly earnings of beginning-of-period employees
W3	num	8	Average monthly earnings of full-quarter employees
WH3	num	8	Average monthly earnings of new hires to full-quarter status
DELTA_B	num	8	Fuzzed: Beginning-of-period employment
DELTA_E	num	8	Fuzzed: End-of-period employment
DELTA_M	num	8	Fuzzed: Employment any time during the period
DELTA_F	num	8	Fuzzed: Full-quarter employment
FPV	num	8	Full-quarter employment - previous quarter
DELTA_FPV	num	8	Fuzzed: Full-quarter employment - previous quarter
A	num	8	Accessions
DELTA_A	num	8	Fuzzed: Accessions
CA	num	8	Accession into Continuing Quarter Employment
DELTA_CA	num	8	Fuzzed: Accession into Continuing Quarter Employment
FA	num	8	Flow into full-quarter employment
DELTA_FA	num	8	Fuzzed: Flow into full-quarter employment
S	num	8	Separations
DELTA_S	num	8	Fuzzed: Separations
CS	num	8	Separations from Continuing Quarter Employment
DELTA_CS	num	8	Fuzzed: Separations from Continuing Quarter Employment
FS	num	8	Flow out of full-quarter employment
DELTA_FS	num	8	Fuzzed: Flow out of full-quarter employment
FSNX	num	8	Flow out of full-quarter employment - next quarter
DELTA_FSNX	num	8	Fuzzed: Flow out of full-quarter employment - next quarter
H	num	8	New hires
DELTA_H	num	8	Fuzzed: New hires
CH	num	8	New Hires into Continuous Quarter Employment
DELTA_H3	num	8	Fuzzed: Full-quarter new hires
R	num	8	Recalls
DELTA_R	num	8	Fuzzed: Recalls
CR	num	8	Recalls into Continuous Quarter Employment
DELTA_W1	num	8	Fuzzed: Total payroll of all employees
DELTA_W2B	num	8	Fuzzed: Average monthly earnings of beginning-of-period employees
DELTA_W3	num	8	Fuzzed: Average monthly earnings of full-quarter employees
WFA	num	8	Average monthly earnings of transits to full-quarter status
DELTA_WFA	num	8	Fuzzed: Average monthly earnings of transits to full-quarter status
DELTA_WH3	num	8	Fuzzed: Average monthly earnings of new hires to full-quarter status

continues on next page

Table 15 – continued from previous page

Variable Name	Type	Length	Description
WFS	num	8	Average monthly earnings of separations from full-quarter status
DELTA_WFS	num	8	Fuzzed: Average monthly earnings of separations from full-quarter status
EBAR	num	8	Average employment
DELTA_EBAR	num	8	Fuzzed: Average employment
JF	num	8	Net job flows
DELTA_JF	num	8	Fuzzed: Net job flows
JC	num	8	Job creation
DELTA_JC	num	8	Fuzzed: Job creation
JCR	num	8	Average job creation rate
JD	num	8	Job destruction
DELTA_JD	num	8	Fuzzed: Job destruction
JDR	num	8	Average job destruction rate
FBAR	num	8	Average full-quarter employment
DELTA_FBAR	num	8	Fuzzed: Average full-quarter employment
FJF	num	8	Net change in full-quarter employment
DELTA_FJF	num	8	Fuzzed: Net change in full-quarter employment
FG	num	8	Average full-quarter employment growth rate
FJC	num	8	Full-quarter job creation
DELTA_FJC	num	8	Fuzzed: Full-quarter job creation
FJCR	num	8	Average full-quarter job creation rate
FJD	num	8	Full-quarter job destruction
DELTA_FJD	num	8	Fuzzed: Full-quarter job destruction
FJDR	num	8	Average full-quarter job destruction rate
AR	num	8	Average accession rate
SR	num	8	Average separation rate
FAR	num	8	Average rate of flow into full-quarter employment
FSR	num	8	Average rate of flow out of full-quarter employment

CHAPTER

FOUR

PERSON-LEVEL FILES

4.1 Individual Characteristics Files - Demographics

4.1.1 Overview

The Individual Characteristics Files (ICF) family of files contains person specific demographic information. Characteristics include gender, age/date-of-birth (DOB), place-of-birth (POB), race, ethnicity, and education. Completed information is available for all individuals who appear in the wage data (workers).

- **Individual Characteristics File (ICF_US)**

  This is the core table with characteristics for all workers. There is one record for every person on the EHF. Characteristics have been completed using a combination of observed data and imputation.

  - Scope: National
  - Key: PIK

- **Date of Birth, Sex, and Place of Birth Implicates (ICF_US_IMPLICATES_AGE_SEX_POB)**
- **Education Implicates (ICF_US_IMPLICATES_EDUCATION)**
- **Race and Ethnicity Implicates (ICF_IMPLICATES_RACE_ETHNICITY)**

  Workers with incomplete information in the observed data table receive imputed characteristics using a set of missing data models. When a variable is missing, multiple implicates (10 draws from the Posterior Predictive Distribution, or PPD) are provided. The IMPLICATES tables contain one record for each worker missing one or more of the characteristics provided on that table.

  - Scope: National
  - Key: PIK

4.1.2 User Guidance

Imputation Flagging

On the core ICF table (ICF_US), each demographic characteristic for a PIK is reported with an observed value if available, and otherwise is imputed. If imputed data is given for a characteristic, the first implicate from the corresponding IMPLICATES table is used. Each characteristic has an associated impute flag (*[char]_imputed*). The impute flags on the file show whether the data was observed (1) or imputed (2). A flag with a value (3) identifies characteristics where reported information failed internal consistency checks and was imputed. The imputed value is given on the ICF_US table.

Estimated standard errors should be adjusted when using characteristics from the ICF. Please see McKinney *et al.* [2021] for a method of producing standard errors that account for the additional imputation variability.

Observed Demographic Data

Demographic information is received by LEHD from the following data sources:

- **Decennial Census**
  - 2000 short form
  - 2000 long form
  - 2010 short form
- **Census Numident**
  - Edited and collapsed longitudinal transaction data dervied from the Social Security Administration (SSA) Numident
- **American Community Survey (ACS)**
  - 2001-present

Observed data for sex, date of birth, place of birth, race, and ethicity may come from any of the above sources. The level of education is provided only on the 2000 long form and the ACS. The coding systems are reconciled across the data sources and to account for changes over time. The observed data are unduplicated and linked using the person identifier.

Protected Identification Key (PIK) Assignment and Unduplication

While the underlying person identifier is the Social Security Number (SSN), SSNs are replaced with different 9 digit Protected Identification Key (PIK) that maps one to one with a SSN. The Census Numident (as well as wage record data) uses the PIK as the person record identifier by default, however all other data sources have a PIK attached to the person record using the Census Bureau's Person Identificaiton Validation System (PVS). If a record contains a SSN, name, and DOB then the match is first attempted using those three pieces of information. If the information is consistent with the information on the Numident, then the record is verified. However, the survey data sources (ACS and Decennial) typically do not have all three pieces of information required for verification. Instead, a PIK is typically assigned to the survey data record using validation. The primary method for validation is the GeoSearch method. Using adiministrative residence information attached to the Numident, the search for a name and DOB match is limited to SSN records residing at the same physical location as the survey record. This greatly reduces the match-candidates, improving the quality of the match. A general name and DOB search is also performed for the residual records that do not match using the GeoSearch method. Overall match rates are high with at least 90% of survey respondents receiving an assigned PIK. Note that assignment is not exclusive and multiple survey records at a point in time may receive the same PIK. For more information on the PIK process, see Wagner and Layne [2014] and Mulrow *et al.* [2011].

LEHD runs an unduplication process using additional characteristics present on both the Numident and the Survey data, resulting in one record per PIK for each data source-year.

Imputation Models

Each data source (Numident, Decennial, ACS) is the primary input for at least one of the observed demographic characteristics on the ICF_US file. The table below shows the ICF_US variable and the data sources used for each variable. When multiple reports exist, the most recent information is preferred for all characteristics.

Table 1: Data Sources Used for Characteristics in the ICF_US

Characteristic	Source	Percent Missing	Model Stage
Gender	Numident	~5%	A
DOB	Numident	~5%	A
POB	Numident Decennial (2000 and 2010), ACS	~5%	A
Race	Decennial (2000 and 2010), ACS	~20%	B
Ethnicity	Decennial (2000 and 2010), ACS	~20%	B
Education	Decennial (2000), ACS	~85%	C

Gender, DOB, and POB have the smallest amount of missing data (~5%), race and ethnicity have the next highest amount of missing data (~20%), and education has the largest amount of missing data (~85%). The missing data follows a monotone missing data pattern, with one set of records missing all variables, the next set missing race/ethnicity/education, and the final set of records missing only education. The monotone missing data pattern allows us to complete the data in stages, starting from the least missing characteristics and then moving on to the next set of variables, continuing until all chracteristics are complete.

In stage A we complete (replace a missing value with an implicate (the actual value returned from an imputation)) gender, DOB, and POB. With these variables completed we move on to stage B and impute race and ethnicity conditional on the DOB, gender, and POB values from stage A. In the final stage we complete education conditional on the variables imputed in both stage A and B. In addition to the demographic characteristics themselves we also use information from a worker's earnings history and information about the worker's co-workers/co-residents.

We use a Classification and Regression Tree (CART) based approach to cluster groups of similar workers together into cells/bins. Our imputation models are defined by the cells we assign workers to. The more homogeneity of the imputation variable (i.e. education) within a cell, the more predictive power of the model. For example, if almost all of the workers within a cell with observed education have college degrees, the implicates for each worker with missing data would have low variability, resulting in a high quality impute. Ideally, once workers are assigned to cells the variability in education outcomes will be large across cells and small within cells.

The depth of the tree (number of cells) is limited by the amount of reported information for each characteristic. For example, characteristics such as Gender, DOB, and POB have substantially more available information than education. Once the set of cells are determined at each stage, the records with reported data are assigned a cell ID. Within each cell a joint Kernel Density Estimate (KDE) is estimated for the variables that need to be completed at that stage (i.e. gender, DOB, POB in stage A). The estimated joint densities are used to generate implicates for workers with missing data by cell ID by sampling randomly from the density. Implicit in our estimation strategy is the assumption that data is missing at random (MAR) within each cell ID.

The imputation models for each stage are shown below:

Stage A (Gender, DOB, POB; 5% missing data)

Stage A missing data is different than the missing data for the workers that only appear in stage B and C. In principal, every worker with a formal employment arrangement should have an SSN. However, LEHD receives earnings records where the PIK does not exist on the Numident. The vast majority of these workers are likely to be persons residing in the U.S. without a legal right to work (i.e. illegal immigrants). These workers are clearly not MAR, therefore the characteristics of workers in the general population are unlikely to match the characteristics of the workers with missing data. To sidestep the issue we assume that recent legal immigrants have similar characteristics as recent illegal immigrants (the workers with missing data), especially when the worker is employed at firms with a large share of recent legal immigrants.

- ***Cell Variables***

  - Modal Non-Native: If the co-worker fraction native is >0.5 then the modal non-native is set to U.S., else the modal non-native is set to the modal value across all co-workers at all jobs for each worker.
  - Co-Worker Fraction Male: If the fraction male is >0.5 then high, else low.
  - New-Worker: 1 if entered data after the first quarter the state appears, else 0. A worker is more likely to be an immigrant if they first appear after the first year the state reports data.

Stage B (Race, Ethnicity; 20% missing data)

Stage B reported data comes from survey data sources. The workers with missing data are a combination of persons not sampled and non-respondents. The workers with missing race and ethnicity data are much more likely to be MAR than the workers in stage A.

- ***Cell Variables***
  - Gender
  - DOB
  - Bestrace: Collapsed Numident race information
  - Earnings: Average annual earnings across all jobs (quartiles) if bestrace is missing
  - POB_race: Collapsed POB categories based on the modal race of each POB
  - Co-Resident and Co-Worker Black/White, Hispanic/Not-Hispanic groups

Stage C (Education (age 25+); 85% missing data)

Stage C reported data comes from survey data sources. Education data is only used/imputed for workers age 25+. The workers with missing data are a combination of persons not sampled and non-respondents. The workers with missing education data are much more likely to be MAR than the workers in stage A, however, the share of missing data is significantly larger for Stage C than the earlier stages resulting in significantly less observed data available to build a model. For stage C we use two parallel approaches. The first approach uses a log-linear model to impute education based on chracteristics similar to those used for the second non-parametric approach as used above in stages A and B. However, the log-linear approach does not require fully interacted effects across all conditioning variables, providing additional modeling flexibility given the limited reported education data available. The additional modeling flexibility is used to control for smaller geographic groups than the non-parametric approach.

- ***Cell Variables***
  - Gender
  - DOB
  - Race/Ethnicity
  - POB_race_educ: Collapsed POB based on country specific immigrant education levels.
  - Earnings: Average annual earnings across all jobs (ventiles)
  - Earnings FB: Low and high earnings category for foreign born workers
  - Industry: Collapsed NAICS industry groups based on observed education
  - Co-Resident and Co-Worker Black/White, Hispanic/Not-Hispanic groups

4.1.3 Codebook: The ICF_US File

Table Metadata for Individual Characteristics File (ICF_US)

Table 2: Access Requirements for ICF_US

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			X

Description
Demographics, place of birth, and education of workers.

Scope
National

Key
PIK

Sort Order
PIK

Variable Information

Table 3: Variable Information for ICF_US

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
DOB	num	4	Date of birth
SEX	char	1	Gender
POB	char	1	Place of birth
RACE	char	1	Race
ETHNICITY	char	1	Ethnicity
EDUC_C	char	1	Highest educational attainment (age 25+)
SEX_IMPUTED	char	1	Imputation status for sex
RACE_IMPUTED	char	1	Imputation status for race
ETHNICITY_IMPUTED	char	1	Imputation status of ethnicity
EDUC_C_IMPUTED	char	1	Imputation status of educ_c
POB_IMPUTED	char	1	Imputation status for POB
DOB_IMPUTED	char	1	Imputation status for DOB

4.1.4 Codebook: The ICF_US_IMPLICATES_AGE_SEX_POB File

Table Metadata for Date of Birth/Sex/Place of Birth Implicates (ICF_US_IMPLICATES_AGE_SEX_POB)

Table 4: Access Requirements for ICF_US_IMPLICATES_AGE_SEX_POB

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			X

Description
Multiply imputed date of birth, sex, and place of birth variables to complete missing information.

Scope
National

Key
PIK

Sort Order
PIK

Variable Information

Table 5: Variable Information for ICF_US_IMPLICATES_AGE_SEX_POB

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
DOB1-DOB10	num	4	Date of Birth Implicates
SEX1-SEX10	char	1	Gender Implicates
POB1-POB10	char	1	Place of Birth Implicates

4.1.5 Codebook: The ICF_US_IMPLICATES_EDUCATION File

Table Metadata for Education Implicates (ICF_US_IMPLICATES_EDUCATION)

Table 6: Access Requirements for ICF_US_IMPLICATES_EDUCATION

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			X

Description
Multiply imputed education variables to complete missing information.

Scope
National

Key
PIK

Sort Order
PIK

Variable Information

Table 7: Variable Information for ICF_US_IMPLICATES_EDUCATION

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
EDUC_C1-EDUC_C10	char	1	Highest educational attainment (age 25+) Implicates

4.1.6 Codebook: The ICF_US_IMPLICATES_RACE_ETHNICITY File

Table Metadata for Race/Ethnicity Implicates (ICF_US_IMPLICATES_RACE_ETHNICITY)

Table 8: Access Requirements for ICF_US_IMPLICATES_RACE_ETHNICITY

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			X

Description
Multiply imputed race and ethnicity variables to complete missing information.

Scope
National

Key
PIK

Sort Order
PIK

Variable Information

Table 9: Variable Information for ICF_US_IMPLICATES_RACE_ETHNICITY

Variable Name	Type	Length	Description
PIK	char	9	Protected Identification Key
RACE1-RACE10	char	1	Race Implicates
ETHNICITY1-ETHNICITY10	char	1	Ethnicity Implicates

4.2 Individual Characteristics Files - Residential Geography

4.2.1 Overview

This file contains annual records with residential geography for individuals found in the wage data (workers). All records from the source tables are kept for workers, though workers may not have residential geography or jobs in every year. Information is available beginning in 1999. The target reference date for the residence is considered to be April 1 of the year.

- **CPR-Sourced Residence (ICF_US_RESIDENCE_CPR)**

  For 1999-2010, residential geography is sourced from the Composite Person Record (CPR) file. Though the formal CPR was not available in 2011, locations for that year are included in this table using an alternate data source.

  - Scope: National
  - Key: PIK ADDRESS_YEAR

- **RCF-Sourced Residence (ICF_US_RESIDENCE_RCF)**

  For 2012 onward, residential geography is sourced from the Residence Candidates File (RCF).

  - Scope: National
  - Key: PIK ADDRESS_YEAR

4.2.2 User Guidance

Composite Person Record (CPR)

The CPR was the original source of residence data used in LEHD. The CPR file series, which begins in 1999, contains fields that provided a linkage between a unique person record and a place of residence for each year. The CPR was produced until the file was discontinued in 2011 (after the production of data year 2010). In the following year, the MAF-ARF (Master Address File-Auxiliary Reference File) was used as a replacement for the CPR for data year 2011. The MAF-ARF was found to differ from the CPR in a number of ways, including a difference in coverage and a lack of deduplication among PIKs. LEHD was able to produce a deduplicated version of the MAF-ARF by defining some very basic business rules. Some additional history of the CPR and MAF-ARF can be found in Graham *et al.* [2017].

The county codes on the CPR (five-digit state+county FIPS code) are contemporaneous. The set of valid county codes and the boundaries of the counties can change over time. For more information on major county changes over time, see the Census Bureau's County Changes page.

Residence Candidates File (RCF)

The RCF combines a set of federal administrative source files containing residence information for a person at a time into a file with preference weights for each person/location by reference period and with no remaining source information. The complete methodology is described in Graham *et al.* [2017].

This extract uses only the most preferred location for each PIK from each year's RCF, so that there is only one record per PIK-year in the ICF.

If additional geographical information is needed (beyond the state, county, and census tract provided in this extract), researchers can lookup a records MAFID in the MAF/TIGER Extract (MAFX). The MAFX is not part of the LEHD Snapshot and must be requested separately for projects. The most recent MAFX will contain the vast majority of MAFIDs found in this extract; however, in some cases researchers may need to find individual MAFIDs in older MAFX files.

4.2.3 Codebook: The ICF_US_RESIDENCE_CPR File

Table Metadata for Residence Geography, 1999-2011 (ICF_US_RESIDENCE_CPR)

Table 10: Access Requirements for ICF_US_RESIDENCE_CPR

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			

Description
Residence geography prior to 2012 for individuals found in the wage data.

Scope
National

Key
PIK ADDRESS_YEAR

Sort Order
PIK

Variable Information

Table 11: Variable Information for ICF_US_RESIDENCE_CPR

Variable Name	Type	Length	Description
PIK	char	9	PIK - Protected Identification Key
ADDRESS_YEAR	num	4	Year address is found on source data
GEOCODEFULL	char	15	FIPS State(2) ||FIPS County(3) ||Tract(6)||Block(4)
LATITUDE_LIVE	num	8	Latitude of residence, 6 implied decimal places
LONGITUDE_LIVE	num	8	Longitude of residence, 6 implied decimal places
FLAG_LATLONG	num	3	Flag quality of latitude/longitude of residence

4.2.4 Codebook: The ICF_US_RESIDENCE_RCF File

Table Metadata for Residence Geography, 2012 forward (ICF_US_RESIDENCE_RCF)

Table 12: Access Requirements for ICF_US_RESIDENCE_RCF

	State Approval Required	IRS Approval Required	SSA Approval Required
Access Requirements			

Description
Residence geography from 2012 forward for individuals found in the wage data.

Scope
National

Key
PIK ADDRESS_YEAR

Sort Order
PIK

Variable Information

Table 13: Variable Information for ICF_US_RESIDENCE_RCF

Variable Name	Type	Length	Description
PIK	char	9	Protected Indentification Key PIK
ADDRESS_YEAR	num	4	Year address is found on source data
COUNTY_LIVE	char	5	Current state/county FIPS as of year indicated in year variable - concatenates CURSTATE and CURCOUNTY
TRACT	char	6	Current Census Tract
MAFID	char	9	MAFID from best source
LATITUDE_LIVE	num	8	Internal point latitude, 6 implied decimals
LONGITUDE_LIVE	num	8	Internal point longitude, 6 implied decimals
FLAG_RCF	num	3	Flag for geographic precision

CHAPTER

FIVE

ACRONYMS USED

BR
Business Register, formerly known as the SSEL

BRB
Business Register Bridge

BLS
Bureau of Labor Statistics

CBSA
Core-Based Statistical Area

CPR
Composite Person Record. A source of residence prior to 2011.

DRB
Disclosure Review Board

ECF
Employer Characteristics File

EHF
Earnings History File

EIN
(Federal) Employer Identification Number

ES202
Historic reference for the employer data provided for the Quarterly Census of Employment and Wages (QCEW). The “ES” mnemonic appears in variable names and data values to indicate derivation from this source.

FIPS
Federal Information Processing Standards

FSRDC
Federal Statistical Research Data Center

FTI
Federal Tax Information. This term describes data that derive from Internal Revenue Service (IRS) information. It is also referred to as Title 26, or T26.

GAL
Geocoded Address List.

ICF
Individual Characteristics File. This family of tables collects observed and imputed person-level demographic characteristics.

J2J
Job-To-Job Flows

LBD
Longitudinal Business Database.

LED
Local Employment Dynamics. This term is used to refer to the voluntary partnership between the U.S. Census Bureau and state entities. The public use data products are considered LED products (QWI, J2J, LODES, PSEO, etc.).

LEHD
Longitudinal Employer Household Dynamics. This refers to the data infrastructure assembled within Census to generate the LED data products. The LEHD Snapshot is a product of that data infrastructure. The term can also refer to the program within the Center for Economic Studies at Census that manages this infrastructure.

LMI
Labor Market Information. These are state government offices responsible for collecting and disseminating data on the labor market within the states. Most have joined the LED partnership with Census, and provide some of the core data of the LEHD infrastructure.

LODES
LEHD Origin-Destination Employment Statistics. This is the public use data product providing information on linkages between place of residence and employment for jobs. This forms the microdata for the OnTheMap application.

MAF
Master Address File. An internal frame of residences and associated geographic characteristics.

MAF-ARF
Master Address File - Administrative Records File. A source of residence information in 2011.

MAFX
Master Address File Extract. An extract of the MAF that can be requested by researchers.

MOU
Memorandum of Understanding. These are agreements between agencies that govern data sharing and usage of that data.

MSA
Metropolitan Statistical Area

OPM
Office of Personnel Management. OPM partners with LEHD to provide job information for federal employees. This data is not part of the LEHD Snapshot.

OTM
OnTheMap. This the public use LED application displaying places of work and residence for jobs.

PIK
Protected Identification Key. This is an encrypted Social Security number used within Census and at the FSRDCs for linking person-level data from multiple sources. See Wagner and Layne [2014] and Mulrow *et al.* [2011] for more information.

PSEO
Post-Secondary Employment Outcomes. This is a public use LED data product reporting earnings, location, and industry of employment after completion of post-secondary education.

QCEW
Quarterly Census of Employment and Wages. The microdata for this BLS public use data product is collected by state LMI offices and shared with Census through the LED partnership.

QWI
Quarterly Workforce Indicators

RCF
Residence Candidate File. A source for residence information after 2011.

RDC
Research Data Centers. Predecessor to the FSRDC system.

SEIN
State employer identification number. It is constructed from the state numeric (*FIPS*) code and the UI account number. The BLS refers to the UI account number in combination with the reporting unit number as SESA-ID.

SEINUNIT
SEIN reporting unit. This is the place of work (establishment) reported on the QCEW.

SSEL
Standard Statistical Establishment List

SPF
Successor-Predecessor File

SSN
Social Security Number

U2W
Unit-to-Worker Impute. This is an imputation procedure to assign establishments (SEINUNITs) to jobs at multi-establishment SEINs in the LEHD infrastructure.

UI
Unemployment Insurance. In the LEHD Infrastructure, UI wage records are the quarterly earnings reports that form the jobs data. These records are used in the administration of the state unemployment insurance system.

VEO
Veteran Employment Outcomes. This is a public use LED data product reporting earnings outcomes after military service.

CHAPTER

SIX

CODEBOOK VALUE LABELS

Categorical variable value labels for this release may be accessed on the LEHD documentation website https://lehd.ces.census.gov/data/lehd-snapshot-doc/S2021/.

BIBLIOGRAPHY

[ASV+09] John M Abowd, Bryce E Stephens, Lars Vilhuber, Fredrik Andersson, Kevin L McKinney, Marc Roemer, Simon Woodcock, and others. *The LEHD infrastructure files and the creation of the Quarterly Workforce Indicators*, pages 149–230. Volume 68. University of Chicago Press Chicago, 2009. URL: http://www.nber.org/chapters/c0485.pdf.

[ASV06] John M. Abowd, Bryce E. Stephens, and Lars Vilhuber. Confidentiality protection in the Census Bureau's Quarterly Workforce Indicators. Technical Paper TP-2006-02, LEHD, U.S. Census Bureau, 2006. URL: http://econpapers.repec.org/paper/centpaper/2006-02.htm.

[BHLM07] Gary Benedetto, John Haltiwanger, Julia Lane, and Kevin McKinney. Using worker flows in the analysis of the firm. *Journal of Business and Economic Statistics*, 25(3):299–313, July 2007.

[CFG+21] Melissa Chow, Teresa C. Fort, Christopher Goetz, Nathan Goldschlag, James Lawrence, Elisabeth Ruth Perlman, Martha Stinson, and T. Kirk White. Redesigning the Longitudinal Business Database. Working Papers 21-08, Center for Economic Studies, U.S. Census Bureau, May 2021. URL: https://ideas.repec.org/p/cen/wpaper/21-08.html.

[GKS17] Matthew R Graham, Mark J Kutzbach, and Danielle H Sandler. Developing a residence candidate file for use with employer-employee matched data. *US Census Bureau, Center for Economic Studies Working Paper 17-40*, 2017. URL: https://ideas.repec.org/p/cen/wpaper/17-40.html.

[HHM+14] John Haltiwanger, Henry Hyatt, Erika McEntarfer, Liliana Sousa, and Stephen Tibbets. Firm Age And Size In The Longitudinal Employer-Household Dynamics Data. Working Papers 14-16, Center for Economic Studies, U.S. Census Bureau, March 2014. URL: http://ideas.repec.org/p/cen/wpaper/14-16.html.

[HMA+17] Samuel Haney, Ashwin Machanavajjhala, John M Abowd, Matthew Graham, Mark Kutzbach, and Lars Vilhuber. Utility cost of formal privacy for releasing national employer-employee statistics. In *Proceedings of the 2017 ACM International Conference on Management of Data*, 1339–1354. 2017. doi:10.1145/3035918.3035940.

[MKA+08] Ashwin Machanavajjhala, Daniel Kifer, John Abowd, Johannes Gehrke, and Lars Vilhuber. Privacy: theory meets practice on the map. In *2008 IEEE 24th international conference on data engineering*, 277–286. IEEE, 2008. doi:10.1109/ICDE.2008.4497436.

[MV11a] Kevin McKinney and Lars Vilhuber. Lehd infrastructure files in the census rdc: overview of s2004 snapshot. Working Papers 11-13, Center for Economic Studies, U.S. Census Bureau, April 2011. URL: http://ideas.repec.org/p/cen/wpaper/11-13.html.

[MV11b] Kevin McKinney and Lars Vilhuber. LEHD Data Documentation LEHD-OVERVIEW-S2008-rev1. Working Papers 11-43, Center for Economic Studies, U.S. Census Bureau, December 2011. URL: http://ideas.repec.org/p/cen/wpaper/11-43.html.

[MGVA21] Kevin L McKinney, Andrew S Green, Lars Vilhuber, and John M Abowd. Total error and variability measures for the quarterly workforce indicators and lehd origin-destination employment statistics in onthemap. *Journal of Survey Statistics and Methodology*, 9(5):1146–1182, 2021. doi:10.1093/jssam/smaa029.

[MMPF11] Edward Mulrow, Ali Mushtaq, Santanu Pramanik, and Angela Fontes. Assessment of the u.s. census bureau's person identifiation validation system. Technical Report, NORC at the University of Chicago, March 2011. URL: https://www.norc.org/Research/Projects/Pages/census-personal-validation-system-assessment-pvs.aspx.

[Vil18] Lars Vilhuber. LEHD Infrastructure S2014 files in the FSRDC. Working Papers 18-27, Center for Economic Studies, U.S. Census Bureau, September 2018. URL: https://ideas.repec.org/p/cen/wpaper/18-27.html, doi:.

[WL14] Deborah Wagner and Mary Layne. Person identification validation system (pvs): applying the center for administrative records research and applications' record linkage software. Working Papers 2014-01, Center for Administrative Records Research and Applications Working Paper. US Census Bureau, 2014. URL: https://www.census.gov/content/dam/Census/library/working-papers/2014/adrm/carra-wp-2014-01.pdf.